



VIT
CHENNAI



CodeCrafters



IEEE
Solid-State
Circuits Society



24-HOURS HACKATHON

HACKTRONICS

2ND EDITION

PROBLEM
STATEMENTS

Contents

Six domains · Sixteen problem statements

01	Artificial Intelligence & Machine Learning	3
02	Healthcare	4
03	Internet of Things	5
04	Autonomous Vehicles & Navigation	6
05	Disaster & Emergency Response	7
06	Open Innovation	8

Artificial Intelligence & Machine Learning

Three problem statements. Each is a modelling problem with a numeric answer — there is no text to prompt a language model with, and no submission is complete without a model that was actually trained.

AI-01

Turbofan Engine Remaining Useful Life Prediction

Given multivariate sensor readings taken from a jet engine across its operating life, predict how many operating cycles remain before it fails. Sensors degrade differently across engine units and across a single engine's own lifetime, and the relationship between a raw reading and cycles-to-failure is neither obvious nor linear.

WHAT WE EXPECT

- A baseline regression model predicting remaining useful life per engine unit.
 - A second, sequence-based model — LSTM or GRU — trained alongside it. A single model with no sequence comparison is not a complete submission.
 - A plotted predicted-versus-actual curve on held-out test units.
 - Performance reported as RMSE against true remaining useful life.
 - The team able to explain why the sequence model's error differs from the baseline's.
-

AI-02

Electricity Theft Detection from Smart Meter Data

Given several years of daily household electricity consumption, flag the customers whose usage pattern suggests meter tampering or theft. Confirmed theft accounts for a small fraction of customers and roughly a quarter of daily readings are missing, so a model that predicts "no theft" everywhere will score well and be worthless.

WHAT WE EXPECT

- A classifier that outputs a confidence score per account, not only a hard label, together with a ranked list of the most suspicious accounts.
 - Performance reported as AUC-PR. Plain accuracy will be treated as a non-answer.
 - A temporal train and test split — earlier days to train, later days to test — with the team able to explain why a random split leaks information here.
 - Missing readings explicitly handled rather than silently dropped.
 - At least one engineered feature beyond the raw daily values, such as rolling statistics, weekly aggregates or day-of-week structure.
 - A stated and defended approach to class imbalance.
-

AI-03

Unsupervised Anomalous Sound Detection in Industrial Machines

Given audio of a machine running normally — and only normally — decide whether a new recording is normal or anomalous. No faulty examples exist at training time, which rules out an ordinary classifier: the model must learn what normal sounds like well enough that a fault stands out against it, on a factory floor where the machine is mixed with real background noise.

WHAT WE EXPECT

- A model trained on normal audio alone. Any use of anomalous clips in training or threshold fitting must be declared, and forfeits the unsupervised claim.
- An explicit time-frequency front end such as log-mel spectrograms, not raw waveform samples fed into a dense network.
- A continuous anomaly score per clip, with the decision threshold chosen and justified separately from the score.
- Performance reported as AUC and partial AUC, computed per machine identifier rather than pooled.
- At least two machine identifiers evaluated, and five clips chosen live by the judges scored on the spot. The team must be able to explain why the score separates well on one identifier and poorly on another.

Healthcare

Three problem statements drawn from clinical and traditional-medicine settings. Interpretability carries as much weight as raw accuracy: in every case the team must be able to say what drove a given output.

HC-01

Fetal Distress Detection from CTG Signals

Build a classifier that reads fetal heart-rate and uterine-contraction signals from cardiotocography recordings and labels the fetal state as normal, suspect or pathological. The classes are imbalanced and the errors are consequential in both directions — a missed distress and a false alarm each carry real cost.

WHAT WE EXPECT

- A trained classifier evaluated on a held-out split.
 - A confusion matrix across all three classes.
 - At least two different model families trained and compared. A single model with no comparison is not a complete submission.
 - A feature-importance analysis explaining which features drove each model's predictions.
 - Performance reported as macro F1. Accuracy alone is not accepted.
-

HC-02

ICU Patient Deterioration Early Warning

Using only the first forty-eight hours of a patient's ICU vitals and laboratory measurements, predict the probability of in-hospital mortality. The data is irregularly sampled and heavily incomplete, and how that missingness is handled changes what the model actually learns.

WHAT WE EXPECT

- A per-patient risk probability, not a binary label alone.
 - A calibration plot and an ROC plot on a held-out test set.
 - Performance reported as AUC-ROC.
 - At least two imputation strategies implemented and compared. One approach with no comparison is not a complete submission.
 - The team able to explain how the choice of imputation moved the result.
-

HC-03

Ayurvedic Herb and Symptom Matcher

Build a conversational system that takes reported symptoms, asks branching follow-up questions to disambiguate them, and returns a ranked list of Ayurvedic herbs or formulations with the reasoning behind each. Every herb-to-symptom mapping must be traceable to a named classical source; undocumented mappings are disqualified.

WHAT WE EXPECT

- A structured knowledge base, submitted as a file alongside the system, covering at least thirty herbs across at least eight symptom categories. Each entry must name its classical source, indications, contraindications and incompatible combinations.
- Follow-up questions that branch on prior answers. A flat questionnaire that asks everything regardless of input does not qualify.
- Ranked recommendations, with the basis of the ranking explained.
- Contraindications checked before anything is returned. A contraindicated herb must never appear in the output.
- At least one dietary or lifestyle recommendation alongside the herbs.
- Three live sessions: clean input, contradictory input and incomplete input. On contradictory input the system must detect the conflict, surface it, ask a tiebreaker question and resolve it with reasoning. Silently returning a recommendation fails that session outright.

Internet of Things

Three problem statements about systems that stay correct when their inputs do not. Evaluation is live and adversarial: judges will drive the running system themselves.

APPLIES TO ALL THREE STATEMENTS IN THIS DOMAIN

- No polling anywhere in the live data path. The pipeline must be push-based, WebSocket or SSE, end to end.
- Sustained throughput of 500 events per second at under 200 ms of interface latency.
- Judges control the mock generator's seed at demo time. Teams will know the class of faults in advance, never the exact sequence.
- Two judges will operate the running system simultaneously from separate sessions, performing different actions at once, with no state corruption and no desynchronisation between views.

IOT-01

Fusion Dashboard with Fault-Resilient Alert Pipeline and Live Replay

Ingest feeds from several simulated sensors with unknown fields, differing formats and differing rates; fuse them into unified live charts with threshold alerts; and make the full pipeline — feed, rule check, alert, alert-clear — survive corrupted input while remaining exactly reproducible on replay.

WHAT WE EXPECT

- Live charts, with alerts that fire and clear correctly against defined thresholds.
- A pipeline that never crashes on NaNs, wrong types, duplicate timestamps or out-of-order data.
- Replay of the last sixty seconds of raw and corrupted input from a persisted log.
- Replay output identical to the original live run. This means genuine event sourcing, not live patching of bad values.

IOT-02

Adaptive Fleet Health Monitoring with Concurrent Session Coordination

Monitor a fleet of fifty devices using per-device baselines the system learns rather than static thresholds, classify live risk status, detect cross-device conflicts within a region, and hold state consistent across simultaneous dashboard sessions.

WHAT WE EXPECT

- Correct classification of all five injected failure modes: drift, spike, flatline, oscillation and sensor swap.
- Baselines that genuinely adapt per device. A hardcoded or globally shared threshold does not qualify.
- Cross-device conflict detection within a region, with the affected devices spotlightable on selection.
- State changes made in one view propagating to every other open view in real time.
- Duplicate device identifiers detected and resolved when two fleets merge live, with neither device's history lost or overwritten.

IOT-03

Rule Engine with Chaining, Conflict Arbitration and Cycle Detection

Build an interface for defining if-this-then-that automation rules against a simulated environment, supporting chained rules, priority-based arbitration when two rules contradict each other, and live detection of circular dependencies.

WHAT WE EXPECT

- A rule engine modelled as a dependency graph, not a flat list of conditionals evaluated in definition order.
- Rules that fire only when their condition is genuinely true, and chains that execute in the correct order.
- Contradictory rules resolved through a mechanism the interface makes visible. A judge must be able to see why one rule won.
- A circular dependency injected by a judge into the running system detected and reported clearly, never hanging, looping or crashing.
- New rules added live, with no restart of the system.

Autonomous Vehicles & Navigation

Three problem statements sitting in the gap between sensing and deciding. In each case the system must not only produce an answer but state how confident it is and why.

AV-01

Intelligent Navigation and Decision-Support System

Convert real-time or simulated sensor detections into explainable driving guidance. The system must not merely register that something is present, but reason about what that presence means for the vehicle's next move, and justify that reasoning to a human observer at any point in the trip.

WHAT WE EXPECT

- An environmental awareness module producing hazard events with position and risk level.
 - A guidance module converting those events into a recommended action, with stated reasoning.
 - A live visual interface showing the vehicle, the detected hazards and the current recommendation.
 - A performance log capturing at least two or three trip metrics, such as trip time, path deviation or hazard alerts raised.
 - One scenario run end to end, plus graceful handling of at least one edge case such as a sensor gap or two conflicting detections.
 - A short written summary of the decision logic for the judges to read alongside the demonstration.
-

AV-02

Smart Scheduling and Route Management for City Bus Networks

Build a system that treats crew-to-bus assignment and route planning as two views of one operational picture rather than two separate tools. It must automate linked and unlinked duty assignment and detect overlap between a newly proposed route and the routes already running.

WHAT WE EXPECT

- Scheduling that supports both linked and unlinked duty and tracks mandated rest periods.
 - A route map showing existing routes, with new routes drawable and plotted.
 - Automatic overlap and conflict detection between a new route and the existing network.
 - A single dashboard where schedule and route data are viewable together, with linked and unlinked assignments visually distinct at a glance.
 - A summary view generated from the underlying data, such as crew utilisation or route coverage.
 - A defined fallback for at least one conflict scenario, such as no rested crew being available.
-

AV-03

Map-Matching for Highway versus Service Road Classification

Classify a vehicle's position as being on a highway or on its parallel service road under intermittent, noisy or biased GNSS data, and state how confident that call is. Nearest-road snapping fails here by construction: two roads a few metres apart are near-equidistant from any given point.

WHAT WE EXPECT

- A trajectory view plotting a GNSS path against two parallel road options.
- Segment-by-segment classification, with a machine-learning component contributing to it and a stated confidence or accuracy measure.
- An accuracy report comparing classifications against known ground truth on synthetic test trajectories.
- A demonstration under injected GPS noise and missing points.
- Correct behaviour through at least one extended outage of thirty seconds or more, without the classification pipeline breaking.

Disaster & Emergency Response

Three problem statements where the output is a decision someone has to act on under time pressure. A plausible-looking answer with nothing behind it is worse here than no answer at all.

DER-01

Citizen-Reported Disaster Damage Mapping

Take geotagged photographs of disaster hazards submitted by the public, classify each by hazard type and estimated severity, and plot them on a live response map. The input is exactly what it sounds like: ground-level images shot by untrained people under stress, in poor light, at odd angles.

WHAT WE EXPECT

- A trained image classifier sorting submissions into at least three distinct hazard categories.
 - A working capture flow. A web form simulating photo and GPS submission is sufficient; a shipped mobile application is not required.
 - A live map dashboard plotting the classified, geotagged reports.
 - At least five images chosen live by the judges — not the team's own demonstration images — classified on the spot, with the team explaining which visual features drove each call.
-

DER-02

Threat-Zone Estimation for Industrial Fire and Explosion Response

Given a storage facility's location, its tank geometry and volume, and prevailing wind conditions, compute and visualise a graded hazard zone showing where thermal radiation and blast overpressure would endanger responders, so that a safe approach direction can be chosen. There is no dataset here: this is a physics and computational geometry problem.

WHAT WE EXPECT

- An implemented thermal radiation and, or, blast overpressure falloff model the team can defend.
 - Wind-dependent zone geometry: zones elongating downwind and compressing upwind.
 - At least three graded severity bands, not a single boundary.
 - The zones overlaid on an actual map.
 - Two different facility configurations demonstrated, with the team able to explain why the resulting zones differ. A fixed-radius circle drawn on a map is not a solution to this problem.
-

DER-03

Predictive Warning System for Avalanche-Prone Terrain

Score avalanche risk for a given day and region from historical occurrence records and forecast danger ratings, and issue a warning meaningfully before conditions turn critical. The target is not a static seasonal hazard rating but an updatable, data-driven score.

WHAT WE EXPECT

- A trained risk model producing a graded risk level, not a binary flag.
- A temporal train and test split — earlier winters to train, later winters to test — with the team able to explain why a random split is unrealistic for a deployed early-warning system.
- A dashboard visualising risk across time and region.
- A simulated alert-dissemination step. Real SMS or push infrastructure is not required.

Open Innovation

One open track for teams whose idea does not sit inside any of the five domains above, or sits across several of them. The problem is yours to choose; the bar is the same.

OI-01

Bring Your Own Problem

Define and solve a problem of your own choosing. It must be a real problem: you should be able to name who suffers from it today and what they currently do instead. Twenty-four hours must be enough to build something that demonstrably works, rather than something that merely demonstrates. A submission that would sit comfortably inside one of the five domains above should be entered there instead of here.

WHAT WE EXPECT

- The problem stated in one or two sentences, with a named user or setting. A technology in search of an application does not qualify.
- A working artefact that runs live in front of the judges. A slide deck, a design mockup or a recorded walkthrough is not a submission.
- A clear account of what already exists in this space and what your approach does differently. A claim that nothing like it exists will be checked.
- The core of the solution built during the event. Pre-existing work may be used as a foundation, but it must be declared at submission, and the twenty-four-hour contribution must be separable and substantial.
- Scope narrow enough to finish. A working slice of a large idea will always beat a broken version of the whole of it.
- The team able to open the code and explain any part of it on request.
- One honest statement of a failure mode the team knows about, and what it would take to fix.

Teams entering Open Innovation are asked to register their chosen problem with the organising committee before work begins, so that the track can be judged against a stated intent rather than a finished artefact alone.